

Practical 9

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In [1]: # Import necessary libraries
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split, cross_val_score
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report, roc_auc_score
from sklearn.datasets import load_iris
from sklearn.preprocessing import LabelBinarizer
from sklearn.model_selection import cross_val_score

# Load the Iris dataset
iris = load_iris()
X = iris.data
y = iris.target

# Split the dataset into training and testing sets (80% training, 20% testing)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Initialize the Decision Tree Classifier
dt_classifier = DecisionTreeClassifier(random_state=42)

# Train the model
dt_classifier.fit(X_train, y_train)

# Make predictions on the test set
y_pred = dt_classifier.predict(X_test)

# Evaluate the model
accuracy = accuracy_score(y_test, y_pred)
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conf_matrix = confusion_matrix(y_test, y_pred)

# Print confusion matrix
print("Confusion Matrix:")
print(conf_matrix)

# Calculate metrics for each class
# Accuracy
print("\nAccuracy: ", accuracy)

# Error
error = 1 - accuracy
print("Error: ", error)

# Classification report
print("\nClassification Report:")
print(classification_report(y_test, y_pred))

# Perform k-fold cross-validation (let's use 5-fold cross-validation)
cross_val_accuracy = cross_val_score(dt_classifier, X, y, cv=5, scoring='accuracy')

# Print the k-fold cross-validation results
print("\nCross-Validation Accuracy Scores (5 folds):", cross_val_accuracy)
print("Average Cross-Validation Accuracy: ", np.mean(cross_val_accuracy))

# AUC - multi-class AUC using One-vs-Rest (OvR) strategy
# For multi-class AUC, we need to binarize the labels
y_test_bin = LabelBinarizer().fit_transform(y_test)
y_pred_bin = LabelBinarizer().fit_transform(y_pred)
auc = roc_auc_score(y_test_bin, y_pred_bin, average='weighted', multi_class='ovr')

# Print the AUC score
print("\nAUC Score: ", auc)
```

Confusion Matrix:

```
[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]
```

Accuracy: 1.0

Error: 0.0

Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	10
1	1.00	1.00	1.00	9
2	1.00	1.00	1.00	11
accuracy			1.00	30
macro avg	1.00	1.00	1.00	30
weighted avg	1.00	1.00	1.00	30

Cross-Validation Accuracy Scores (5 folds): [0.96666667 0.96666667 0.96666667 0.93333333 1.0]

Average Cross-Validation Accuracy: 0.9533333333333334

AUC Score: 1.0